Drone Prototype Comparison

Ten drones we could 3D print. Which two are the best start for our prototype?

Name: _____ **Date:** _____

Our solution, updated after Bob Leigh's advice

Satellites show where the invasive grass grows. Then **two drones work together**. Bob told us: "The two drones could work together."

🔍 Scout drone
Light and fast. Flies over the grassy areas with a **heat (infrared) camera** and **LIDAR** to find small fires early. Needs: **speed**, **distance**, and room for light sensors.

💧 Water drone
Bigger and slower. A person checks the scout's video, then sends this drone to drop water on the fire. Needs: **carrying capacity** and **distance**.

What to do

1. **Look at each drone** on pages 2 to 6: the picture, the facts and what it does.
2. **Circle a score from 1 to 5** for each of the four questions. Use the scoring guide below.
3. **Add up the total** (out of 20), tick its **best job** (scout, water drone or neither), and write what we would change.
4. **Copy your scores** to page 7 and pick your best scout and best water drone. Then we add up the team scores on page 8.

Scoring guide

Score it on …	1 No	3 Some	5 Yes!
Carrying capacity Could it carry water, or a camera and sensors?	No room and no power to lift anything.	Could carry something light, like a small cup of water.	Has room and power to carry a real load.
Speed Could it get to a fire quickly?	Slower than a bike (under 30 km/h), or it mostly hovers.	About as fast as a car on the highway (100 km/h).	Faster than a race car (200 km/h or more).
Distance Could it cover a big area of rainforest?	A few km and back, about 20 minutes in the air.	Tens of km, about an hour in the air.	Hundreds of km, many hours in the air.
Can we build it? With our printer, time and help	Too hard, too big or too costly for us.	We can print it, but need help with motors and electronics.	We can print it and put it together ourselves.

Use 2 and 4 for in between. The highest total is 20.

Score the real, full-size drone

- Score **speed** and **distance** as if it were the **real, working, full-size drone**, not the printed model or a small prototype. For models that don't fly, imagine the design built full size and working.
- Then score **Can we build it?** honestly, including the parts we would have to add.

FLIES really flies **DISPLAY MODEL** doesn't fly as printed **ADD-ON** rides on another drone

1 FP-5 Flamingo (1:35 scale)

DISPLAY MODEL



Winged drone · display model

Print	83 g of plastic, about 5½ hours, 3 plates
Also need	Super glue
Flies?	No. It is a small copy of a real drone.

A small model of a large, long-range winged drone from Ukraine. The real one has long straight wings and a jet engine on its back, and flies very far. This model is glued together and sits on a shelf.

Think: long straight wings help a plane fly far. Could our drone use wings like these?

makerworld.com/en/models/2626260

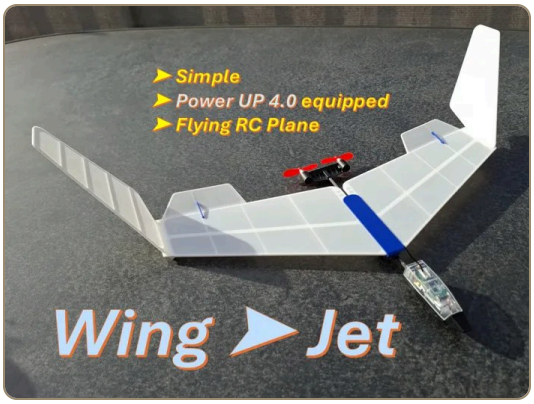
Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

2 WingJet (phone-controlled plane)

FLIES



Flying wing · really flies

Print	29 g, about 2¾ hours, 1 plate
Also need	PowerUp 4.0 kit (motor, propeller, battery), bought separately
Flies?	Yes, outdoors in calm weather. You steer it with a phone app.

A very light flying wing (no tail, just one big wing). The PowerUp kit clips on the front and you fly it from a smartphone. Easy to build and fly, but it is too light to carry much.

Think: could a bigger version carry a small water pod? What happens in wind?

makerworld.com/en/models/1038079

Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

3 World's Fastest Sub-250 g FPV Quad

FLIES



Quadcopter (4 propellers) · really flies

Print	46 g, about 2½ hours, 3 plates
Also need	Flight controller, 4 motors, FPV camera, battery, radio; soldering
Flies?	Yes, very fast. The designer says it breaks if you crash it.

A racing drone shaped like a rocket, built to be as light and fast as possible, like a Formula 1 car. The pilot wears goggles and sees through its camera. Fast, but built light, not strong.

Think: speed is great for getting to a fire, but can it lift anything?

makerworld.com/en/models/458347

Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

4 Delta-wing drone model (listed as "Iranian kamikaze drone")

DISPLAY MODEL



Triangle (delta) wing · display model

Print	7 g, about 1 hour, 1 plate
Also need	Nothing
Flies?	No. Small one-piece model.

A model of a triangle-shaped winged drone with the propeller at the back. Real drones with this shape fly long distances. This one is a quick, small print you can hold in your hand.

Think: is a triangle wing or a straight wing better for carrying a load?

makerworld.com/en/models/2596404

Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

5 Javelin X140 FPV Carrier Aircraft

FLIES



Winged RC plane, 1.4 m wingspan · really flies

Print	1.6 kg of plastic, about 70 hours, 11 plates
Also need	Carbon rods, motor, speed controller, 4 servos, big battery, 6-channel radio
Flies?	Yes. Takes off at 2–2.5 kg. Designer says it is still being tested and is not for beginners.

A big remote-control plane copied from a real drone used in Ukraine. It has a large battery bay, so it can fly far and carry weight. It is the closest to a real fire-fighting winged drone, but it is the hardest and most expensive to build.

Think: could the battery bay or nose hold a water tank or camera instead?

makerworld.com/en/models/2896506

Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

6 NRO4 Nano Rocket FPV drone

FLIES



Tiny quadcopter · really flies

Print	38 g, about 2½ hours, 1 plate
Also need	DJI O4 camera, 2-inch propellers, motors, small 1-cell battery, radio
Flies?	Yes. Fully 3D-printed frame.

A tiny camera drone small enough to fit in your hand. It flies with a small battery and sends live video to goggles. Good for looking around, but too small to carry anything.

Think: could a drone like this be the "scout" that finds the fire or the grass?

makerworld.com/en/models/2175285

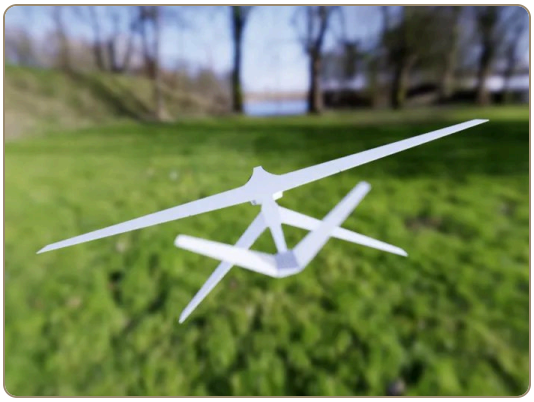
Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

7 Swift Falcon glider

FLIES



Hand-thrown glider · flies, no motor

Print	3–4 g, about 18 minutes, 1 piece
Also need	Nothing
Flies?	Yes, you throw it. Bend the flaps to change how it flies.

A tiny one-piece glider with a main wing, a tail and bendable flaps. It is very tough and fast to print. We can print many and test wing shapes the same day.

Think: test how far it glides, then add a paper clip as a "load". What changes?

makerworld.com/en/models/2467409

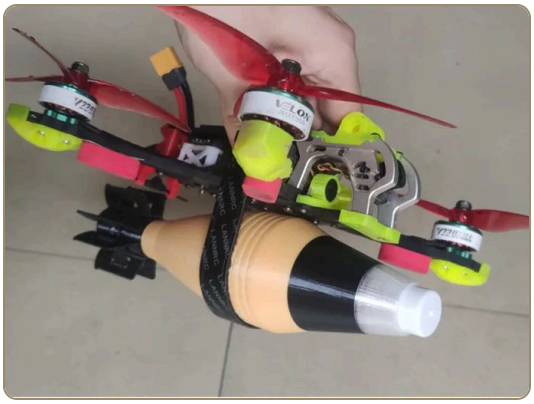
Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

8 Immersive FPV Accessories (drop pod)

ADD-ON



Pod that straps under a quadcopter · add-on

Print	130 g, about 5 hours, 1 plate (pauses to change colours)
Also need	An FPV quadcopter to strap it to, and a strap
Flies?	Not by itself. It rides under another drone.

A pod shaped like a falling bomb with fins, strapped under a racing quadcopter. It is for looks only and does not open. Its shape gives us an idea: a pod that hangs under a drone and could hold water.

Think: how would we make a pod like this open and drop water over a fire?

makerworld.com/en/models/533814

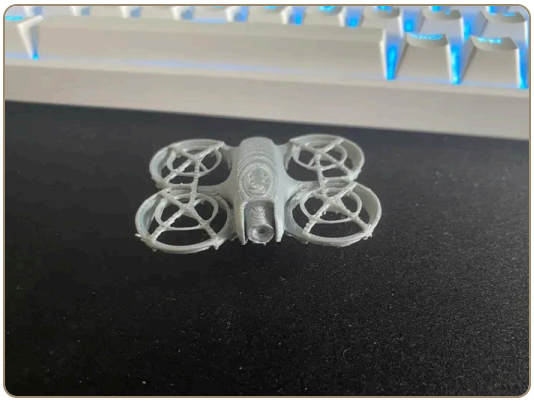
Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

9 DJI Neo model

DISPLAY MODEL



Small camera quadcopter · display model

Print	182 g, about 18 hours, 1 plate
Also need	Nothing (propellers glue on)
Flies?	No. A copy of a real camera drone.

A model of a small real drone with guards around the propellers so it is safe near people. The maker built it from a photo using an AI tool. It can be printed smaller.

Think: propeller guards keep a drone safe. Would our drone need them near trees?

makerworld.com/en/models/2679270

Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

10 Stinger Drone model kit

DISPLAY MODEL



Two-rotor drone with wings · toy kit

Print	15 g, about 45 minutes, 7 parts
Also need	Nothing: snaps together, no glue
Flies?	No. The rotors turn by hand.

A toy drone with two rotors that can tilt, plus small front wings for speed. It mixes a copter and a plane: rotors to lift off, wings to fly fast and far (a "VTOL" drone).

Think: could our drone take off like a copter and fly like a plane?

makerworld.com/en/models/852615

Carrying capacity	Speed	Distance	Can we build it?	Total /20
1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	_____

Best job for this design: ☐ Scout drone *(light, fast, finds hot spots)* ☐ Water drone *(big, carries water)* ☐ Neither

✎ **What would we modify for our project?** (to carry water, map the grass, fly farther ...)

My scores

Name: _____

Copy your circled scores from pages 2 to 6. Add across for the total, out of 20. In the last column, circle **S** if you ticked scout or **W** if you ticked water drone.

Drone	Carrying	Speed	Distance	Can we build it?	Total /20	Best job S or W
1. FP-5 Flamingo (1:35 scale)						**S W**
2. WingJet (phone-controlled plane)						**S W**
3. World's Fastest Sub-250 g FPV Quad						**S W**
4. Delta-wing drone model						**S W**
5. Javelin X140 FPV Carrier Aircraft						**S W**
6. NRO4 Nano Rocket FPV drone						**S W**
7. Swift Falcon glider						**S W**
8. Immersive FPV Accessories (drop pod)						**S W**
9. DJI Neo model						**S W**
10. Stinger Drone model kit						**S W**

✎ **My best scout drone is number ___ because ...** (look at its speed and distance)

✎ **My best water drone is number ___ because ...** (look at its carrying capacity and distance)

Team scores

Each person reads out their total for each drone (from page 7). Add across for the team total, out of 180. Then count how many people circled **S** (scout) and **W** (water drone).

Drone	Sylvie	Brynn	Jack	Fedor	Ira	Jordan	Ella	Alexis	Amelia	Team total	S votes	W votes
1. FP-5 Flamingo (1:35 scale)												
2. WingJet (phone-controlled plane)												
3. World's Fastest Sub-250 g FPV Quad												
4. Delta-wing drone model												
5. Javelin X140 FPV Carrier Aircraft												
6. NRO4 Nano Rocket FPV drone												
7. Swift Falcon glider												
8. Immersive FPV Accessories (drop pod)												
9. DJI Neo model												
10. Stinger Drone model kit												

✓ Our two drones

DESIGN

🔍 **Scout drone:** _____ 💧 **Water drone:** _____

✎ **We picked them because ...** (scout: speed, distance; water drone: carrying, distance; can we build them?)

✎ **How will they work together? Where will we launch them from, and can they fly far enough?** (Bob's question)

✎ **Parts we need to buy or borrow, and who will find them:**